

Kayra Bulbul

+61407338877 | kayrabulldotdev@gmail.com | linkedin.com/in/kayra-bulbul | github.com/KayraBulbul

EDUCATION

RMIT University

Melbourne, VIC

Bachelor of Computer Science, Minors in data science and Enterprise Systems development

Mar. 2024 – Nov 2027

EXPERIENCE

Peer Mentor

July 2026 – Present

RMIT University

Melbourne, VIC

- Mentor students in **Algorithms & Analysis** and **Further Programming**, helping develop understanding of data structures, algorithms, OOP and debugging techniques.
- Guide students through problem-solving approaches and debugging strategies rather than providing direct solutions, strengthening their ability to independently reason about technical problems.

PROJECTS

AgentScope | *Go, SQLite, Docker, OpenAI SDK*

August 2026 – Present

- Designing and building a **transparent AI agent platform** for interacting with LLMs while exposing agent actions, tool usage and execution state to the user.
- Developing an **agent control plane** that enables users to remotely interact with and manage coding agents running on another machine.
- Architecting the backend in **Go** with persistent state in **SQLite**, containerised services with Docker, and integration with the **OpenAI SDK**.

Overworld Atlas | *React, TypeScript, Go, Docker, PostgreSQL*

July 2026 – Present

- Built a **Go backend** service for a Minecraft server I play on with my mates, exposing APIs for server status, authentication and application functionality.
- Implemented the **Minecraft Server List Ping protocol** from scratch, including VarInt encoding, TCP packet construction, handshaking and status-response decoding.
- Added configurable **timeouts, TTL-based caching, CORS, structured logging and HTTP middleware**, with unit and integration tests covering protocol and API behaviour.
- Developing a **React/TypeScript frontend** integrating live server information, an interactive world map and community features with the Go API.

NSW Crime RAG System | *Python, LangChain, ChromaDB, OpenAI, Ollama*

Oct. 2025 – Feb 2026

- Built a **retrieval-augmented generation (RAG) system** for querying NSW crime statistics, using OpenAI embeddings and ChromaDB for semantic retrieval with locally hosted Llama2 inference.
- Implemented document **chunking, vector embedding, persistent vector storage and similarity-based retrieval**, returning source documents alongside generated responses for traceability.
- Developed an **evaluation framework across 10 ground-truth queries**, measuring response effectiveness, faithfulness and source attribution to assess retrieval and generation quality.

SQL Data Warehouse | *T-SQL, SQL Server, ETL, Star Schema*

Feb. 2026 – Feb 2026

- Built an end-to-end **SQL Server data warehouse** using a Bronze–Silver–Gold architecture to integrate CRM and ERP datasets into analytics-ready models.
- Developed stored-procedure ETL pipelines for **bulk ingestion, deduplication, data cleansing, standardisation and validation**, with execution timing and error handling.
- Modelled the Gold layer as a **star schema** with customer and product dimensions and a sales fact table, enabling analysis of customer behaviour, product performance and sales trends.

TECHNICAL SKILLS

Languages: Java, Python, C/C++, SQL, JavaScript/TypeScript, HTML/CSS, Go

Frameworks: React, Node.js, Express.js, TypeORM, FastAPI, Chi

Developer Tools: Git, Docker, VS Code, Neovim, Linux, Power BI, Tableau

Data & Cloud: PostgreSQL, MS SQL Server, Pandas, ETL/ELT, PySpark, Airflow, Snowflake, MongoDB, dbt, AWS